

Markscheme

Mathematics: analysis and approaches
Practice question bank

2. (a)

Consider $a = 1.5 \times 10^2$ and $b = 4 \times 10^1$.

Calculate $C = \sqrt[5]{(ab)^3}$, giving your answer correct to 3 significant figures.

attempt to find the value of ab (M1)

$$ab = (1.5 \times 10^2)(4 \times 10^1) = 6 \times 10^3 \quad \text{A1}$$

attempt to raise to the power 3 and take the 5th root (M1)

$$(ab)^3 = 2.16 \times 10^{11} \quad \text{A1}$$

$$C = \sqrt[5]{2.16 \times 10^{11}} = 185 \text{ (3 s.f.)} \quad \text{A1}$$

Note: Do not accept a final answer given in calculator notation, e.g. $1.85E2$.

[5 marks]

(b)

Hence write $\frac{1}{C}$ in the form $k \times 10^n$, correct to 3 significant figures.

$$\frac{1}{C} = \frac{1}{185} \quad \text{(M1)}$$

$$= 5.41 \times 10^{-3} \quad \text{A1}$$

[2 marks]